Mini-Project – 2B Web based on ML (ITM 601)

PROPOSAL

MINI-PROJECT TITLE

T. E. Information Technology

By

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STASH TONY LOPES	**66**
ANUSHKA KOTAL	**52**
ASHWIN KOONISSERY	**51**

Mentor:

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Mini Project Proposal

Project Title	**EVBharat –** AI-Driven Battery Health Monitoring and Intelligent Route Optimization for Electric Vehicles in India
Project Members (Mention Leader in Bold)	**KARTHIK JANARDHAN** STASH TONY LOPES ANUSHKA KOTAL ASHWIN KOONISSERY
Situation/Problem/Opportunity/Need	Electric Vehicle (EV) adoption in India is limited by range anxiety, battery degradation, inefficient energy usage, and unpredictable traffic and road conditions. Conventional navigation systems do not consider battery health or EV-specific energy consumption. There is a need for an intelligent system that integrates AI, ML, and real-time traffic data to improve EV reliability and user confidence.
Problem Statement	Electric Vehicles (EVs) face challenges related to battery limitations and traffic conditions. This project uses AI and Machine Learning to predict energy consumption, monitor battery health, and recommend energy-efficient routes based on real-time traffic.
Objectives	● Monitor EV battery health (State of Health). ● Predict energy consumption under real driving conditions. ● Integrate real-time traffic and road conditions. ● Determine if current battery is sufficient to reach the destination. ● Recommend optimal routes and driving strategies to improve range efficiency.
Method /Approach (Steps/Modules/Proposed Work/Architectural Dia.)	● Data Collection: Battery, vehicle, traffic, and route data ● Battery Health Estimation using ML models ● Energy Consumption Prediction using regression/ML techniques ● Traffic-aware route analysis using map APIs ● Decision engine for route and battery efficiency recommendations ● Dashboard for visualization and alerts
Success Criteria (Advantages / Performance Metrics)	● ≥90% accuracy in energy consumption prediction ● ≤8% error in battery health estimation ● 10–15% improvement in route energy efficiency ● Reduced range anxiety and improved user confidence

Resources (People ,Time, hardware / software resources, dataset, online survey with goggle form, cost, other)	● People: 4 students ● Time: 8–10 weeks ● Hardware: Laptop/PC ● Software: Python, Pandas, NumPy, Scikit-learn, Flask ● Data: Public EV datasets, simulated telemetry, traffic APIs ● Cost: Low (open-source tools)
Risk and Dependencies	● Limited real-world EV battery data ● Dependence on API availability and data quality ● Model performance depends on dataset realism
Remark (can be continued as BE Project/Outhouse Project)	
References (IEEE Format)	[1] M. Chen, X. Shen, and J. Hu, "Energy consumption prediction of electric vehicles based on real-world driving data," *IEEE Access*, vol. 8, pp. 93322–93336, 2020, doi: 10.1109/ACCESS.2020.2995262. [2] Y. Wang, Z. Sun, and S. Zhang, "Route planning for electric vehicles considering traffic congestion and battery constraints," *IEEE Intelligent Transportation Systems Magazine*, vol. 11, no. 2, pp. 65–77, Summer 2019. [3] A. Neubauer and E. Wood, "The impact of range anxiety and charging infrastructure on electric vehicle adoption," *Transportation Research Part A: Policy and Practice*, vol. 39, no. 2, pp. 102–120, 2019. [4] Google, "Google Maps Platform: Routes and Traffic APIs," Google Developers Documentation, 2024. [Online]. (Accessed: Jan. 2026).